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Experiment 3:
Waveform Generation using Operational Amplifiers

Subject: Electronic Engineering III
Hamm-Lippstadt University of Applied Sciences, Lippstadt
Location: L3.3-E02-080

Scheduled day for the experiment: 15.01.2025 and 16.01.2025
The pre-report must be submitted before conducting the experiment
The final report is due on **January 23, 2025.**

Student Assistance

Office hours should be scheduled with the professor via email, providing the subject of the question in advance.

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Experiment Objectives

- Generate square waves with adjustable frequency using an operational amplifier (OpAmp).
- Generate triangular or sawtooth waves by integrating a square wave signal with the help of an operational amplifier.
- Generate a pulse-width modulation (PWM) signal with a variable duty cycle using an operational amplifier and a comparator circuit.

I. Introduction

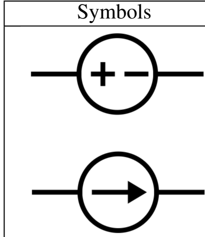
In this experiment, we will explore the use of operational amplifiers (OpAmps) for generating various types of waveforms, such as square waves, triangular waves, and pulse-width modulation (PWM) signals. These waveforms are essential for many applications in electronics, including signal processing and control systems.

The components and devices used in this experiment, including the operational amplifier, potentiometer, voltage source, resistors, capacitors, multimeter, and breadboard, are described in Section II. Detailed instructions for carrying out the experiment and step-by-step guidance on constructing the necessary circuits are provided in Section III.

II. Components and Devices

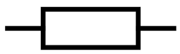
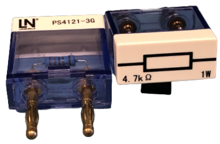
- **Independent Voltage Source and Independent Current Source**
  Independent Voltage Source: A device designed to maintain a fixed voltage across its terminals, regardless of the current flowing through it or the circuit elements connected to it. A common example of this device is a battery.
  Independent Current Source: A device that ensures a specified current flows through itself, independent of the voltage across its terminals or the connected circuit elements. Unlike voltage sources, practical examples of current sources are less common; however, they are valuable for constructing theoretical models. A multimeter can function as both a current and a voltage source in laboratory setups. Table I illustrates the symbols and device examples for independent voltage and current sources.

TABLE I: Symbols and devices for independent voltage and current sources.

Symbols	Device Photo
	

- **Resistor**
  Resistors are passive electrical components that resist the flow of current, dissipating energy as heat. The resistance values of the THT resistors used in this experiment are indicated by the color bands on their surface. Table II presents the symbol and a photograph of the resistors.
  To correctly measure resistance, the equipment must be properly connected. When using a multimeter as an ohmmeter, connect it across the two terminals of the resistor. Ensure that the circuit is powered off during this measurement to avoid inaccuracies or damage to the multimeter. For voltage measurements, a voltmeter should always be connected in parallel with the component. Conversely, for current measurements, an ammeter must be connected in

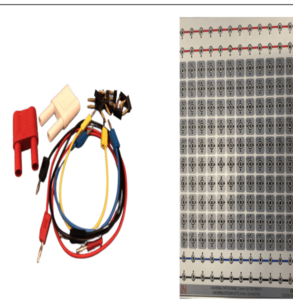
TABLE II: Resistor symbol and device photo.

Symbol	Device Photo
	

series with the component to accurately measure the current flowing through it.

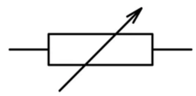
- **Conductor (Wires and Breadboard)**
  A conductor is a material that allows charge to flow through it. It is used to connect circuit elements in an electrical circuit. The conductors commonly are made of metals, as they have low resistance and high conductance. Breadboard, which is a board widely used as a prototype, acts as a conductor and the construction base of electronics. The breadboard allows the quick creation of circuits without making permanent connections. It is a widely used tool to learn electrical circuits. Table III shows the symbol and the photo of the wires and breadboard. Table III shows the symbol and the photo of the wires and breadboard.

TABLE III: Conductor symbol and device photo.

Symbol	Device Photo
	

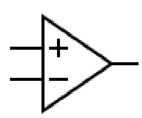
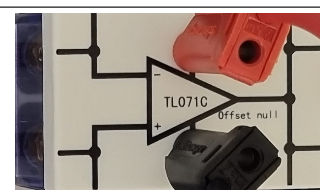
- **Potentiometer**
  A potentiometer is an essential instrument used to measure an unknown voltage by comparing it with a known reference voltage. It operates based on the voltage divider principle, where the unknown voltage is balanced against the known reference voltage. This method allows precise measurements with minimal error. In addition to measuring voltage, a potentiometer can be used to determine the electromotive force (EMF) and internal resistance of a given cell. It is also helpful in comparing the EMF of different cells, providing accurate results without drawing significant current from the sources. The symbol and a photograph of the potentiometer are shown in Table IV. The suggested potentiometer is PS4121-8G.

TABLE IV: Potentiometer symbol and device photo.

Symbol	Device Photo
	

- **Operational Amplifier (OpAmp)**
  An operational amplifier (OpAmp) is a crucial electronic component used to amplify voltage signals. It is widely employed in analog circuits for various functions such as signal conditioning, filtering, and performing mathematical operations, including addition, subtraction, integration, and differentiation. The OpAmp has two inputs: an inverting input, which inverts the phase of the signal, and a non-inverting input, which maintains the original phase of the signal. It is characterized by a high input impedance, which ensures minimal current is drawn from the source, and a low output impedance, enabling effective drive of connected loads. The symbol and a photograph of an operational amplifier are shown in Table V. The suggested operational amplifier is TL071C.

TABLE V: Operational Amplifier symbol and device photo.

Symbol	Device Photo
	

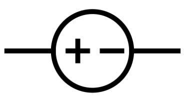
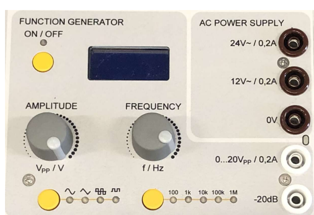
- **Oscilloscope**
  The oscilloscope is an electronic test instrument that graphically displays voltage variations over time. It allows analysis of waveform properties, including amplitude, rise time, and period. Table VI presents the symbol and a photograph of the oscilloscope.
- **Function Generator**
  A function generator produces standard waveforms such as sine, square, and triangular waves. The instrument allows precise control over parameters such as amplitude and frequency, making it essential to test and analyze electronic circuits. Table VII

TABLE VI: Oscilloscope symbol and device photo.

Symbol	Device Photo
	

shows the symbol and a photograph of the function generator.

TABLE VII: Function Generator symbol and device photo.

Symbol	Device Photo
	

III. Experiments

A. Task 1: Generation of a Square Wave Signal

In this task, the operational amplifier (op-amp) circuit is designed to generate a square wave signal. The non-inverting input of the op-amp is connected to a voltage divider formed by resistors R_2 and R_3, where R_2 is a potentiometer. Adjusting R_2 allows for variation in the reference voltage, influencing the threshold levels at which the op-amp switches its output state.

The inverting input is connected to a resistor R_1 and a capacitor C_1. The capacitor undergoes periodic charging and discharging cycles, causing the voltage at the inverting input to oscillate. This oscillation triggers the op-amp to switch states, thereby producing the square wave output. The resistor and capacitor interplay determines the generated signal's frequency.

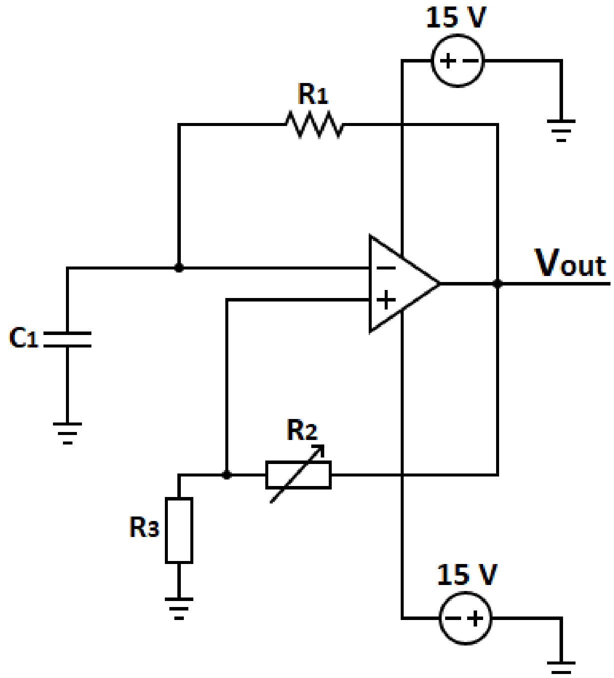


Fig. 1: Square wave generator circuit.

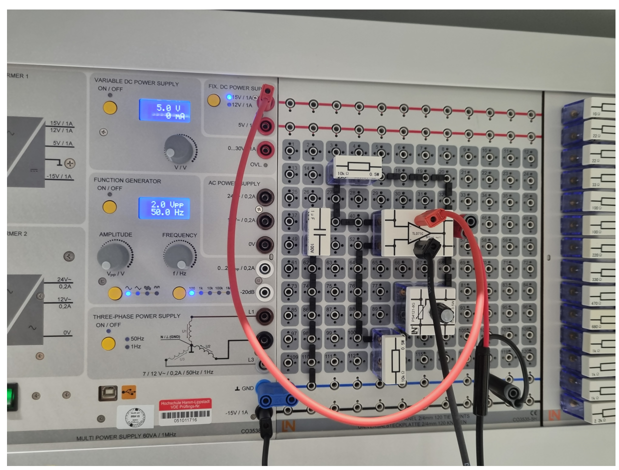


Fig. 2: Assembled square wave generator circuit.

TABLE VIII: Component values for the square wave generator circuit.

Component	Value
V_{out}	Square Wave
R_1	10 kΩ
R_2	10 kΩ
R_3	10 kΩ
C_1	1μF

Instructions

1) Assemble the Square Wave Generator circuit as shown in Figures 1 and 2, using the components listed in Table VIII. Ensure that all connections are

secure and components are correctly placed.

2) Connect one oscilloscope channel to the circuit output. Specifically, attach the probe's input to the circuit's output (V_{out}) and the ground clip to the circuit's ground (GND).
3) Adjust the potentiometer until the output frequency reaches approximately 25 Hz. Observe the waveform on the oscilloscope to verify it resembles a square wave.
4) Measure and record the peak-to-peak voltage (V_{pp}) of the square wave when the frequency is set to 25 Hz.
5) Calculate the expected frequency of the square wave based on the resistor and capacitor values. Use the formula $f = \frac{1}{2 \cdot R_{eq} \cdot C}$, where R_{eq} includes the resistance contribution of the potentiometer.
6) Compare the calculated frequency with the observed frequency. Note any discrepancies and consider possible reasons for these differences.
7) Capture pictures of the oscilloscope display showing the square wave at 25 Hz and include them in your report. Discuss the observed waveform and any deviations from the expected behavior with your colleagues.
8) Write a brief comment on how the capacitance and resistance values influence the frequency of the square wave in the circuit. Include your observations and conclusions in the report.

B. *Task 2: Generation of a Triangular Wave Signal*

The circuit shown in Figures 3 and 4 is connected to the output of a square wave generator. This configuration allows the circuit to integrate the square wave signal. When the square wave is at a positive voltage, the integrator's output decreases linearly over time, producing a downward ramp. Conversely, when the square wave switches to a negative voltage, the integrator's output increases linearly, creating an upward ramp. This behavior occurs because the integrator operates in an inverting configuration, where the polarity of the input signal is reversed in the output. Consequently, the integrator circuit transforms the square wave input into a triangular wave output, providing a distinct and predictable waveform for further analysis or use. Note that the triangular wave frequency is the same as the square wave.

Instructions

1) Assemble the triangular wave generator circuit as shown in Figures 3 and 4. Ensure all connections are secure and components are correctly placed according to the schematic. The input to the circuit is a square wave with a peak-to-peak voltage (V_{pp}) of 4 V and a frequency of 50 Hz.
2) Connect one oscilloscope channel to the input of the circuit (V_{in}). Attach the probe's input to the square wave generator output and the ground clip to the circuit ground (GND).

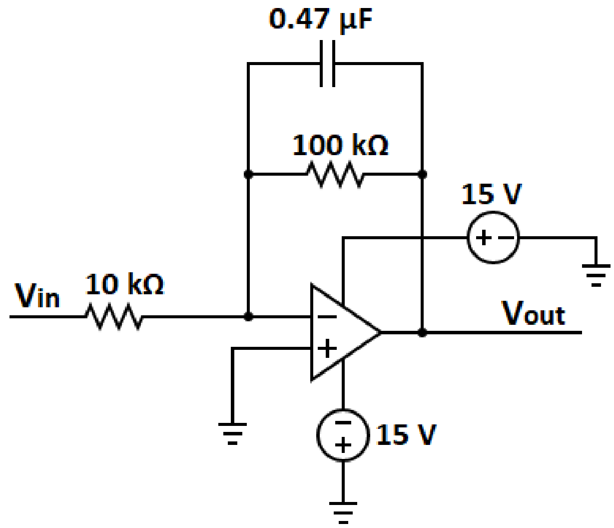


Fig. 3: Circuit for integrating a square wave.

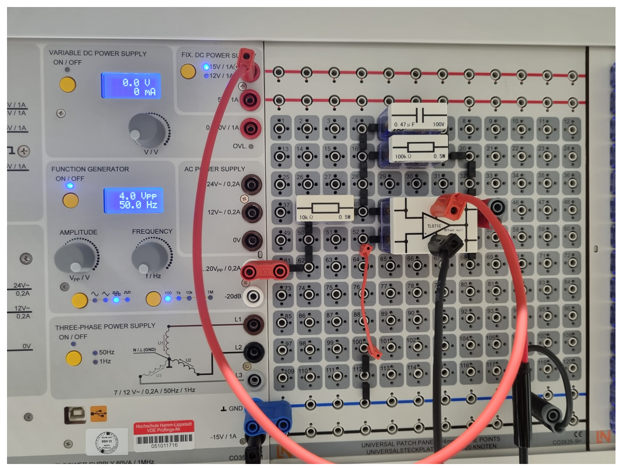


Fig. 4: Assembled circuit for integrating a square wave.

3) Connect another oscilloscope channel to the output of the integrator circuit (V_{out}).
4) Observe the input square wave (V_{in}) and the output triangular wave (V_{out}) simultaneously on the oscilloscope. Record both signals' peak-to-peak voltage (V_{pp}) and frequency.
5) Analyze the relationship between the input and output signals. Note how the square wave is integrated into a triangular wave and how the circuit components affect the output signal's characteristics.
6) Capture clear pictures of the oscilloscope display showing both the input and output waveforms. Include these images in your report.
7) Write a detailed discussion explaining the observed physical phenomenon in your report. Highlight the

role of each component in the circuit and how they contribute to the integration process.

8) Conclude by discussing how the resistor and capacitor values influence the shape and frequency of the triangular wave.

C. Task 3: Generation of a PWM Signal

In this task, the circuit generates a Pulse Width Modulation (PWM) signal using a comparator, as shown in Figures 5 and 6. The comparator receives two input signals: a triangular wave from the previous task and a reference voltage. The comparator continuously compares the triangular wave voltage with the reference voltage.

When the triangular wave voltage exceeds the reference voltage, the comparator outputs a high voltage level (for example, +15V). Conversely, when the triangular wave voltage exceeds the reference voltage, the comparator outputs a low voltage level (for example, -15V). This process produces a PWM signal at the output, where the relationship between the triangular wave and the reference voltage determines the duty cycle. The PWM signal generated by the circuit is widely used in various applications, including motor control, signal modulation, and power management. The ability to control the duty cycle by adjusting the reference voltage demonstrates the flexibility of this method for generating modulated signals.

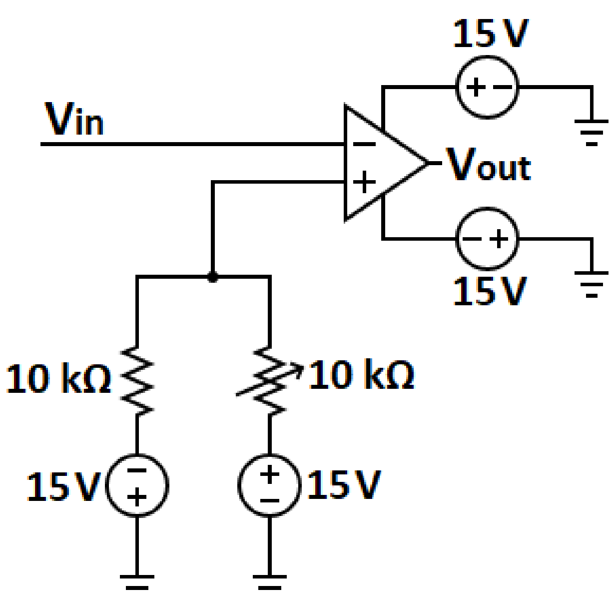


Fig. 5: Circuit for comparing a triangular wave to a reference voltage.

Instructions

1) Assemble the PWM generator circuit as shown in Figures 5 and 6. In this task, the input triangular wave has a peak-to-peak voltage (V_{pp}) of 4 V and a frequency of 50 Hz.

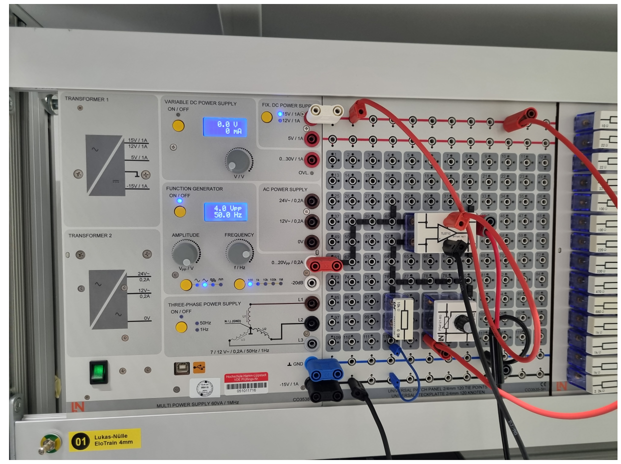


Fig. 6: Assembled circuit for comparing a triangular wave to a reference voltage.

2) Connect the first oscilloscope channel to the input of the circuit (V_{in}):
   - Attach the probe's input (positive terminal) to the output of the triangular wave generator.
   - Connect the probe's ground clip (negative terminal) to the circuit ground (GND).
3) Connect the second oscilloscope channel to the output of the comparator circuit (V_{out}):
   - Attach the probe's input (positive terminal) to the output of the comparator circuit.
   - Connect the probe's ground clip (negative terminal) to the circuit ground (GND).
4) Observe the triangular wave (V_{in}) and the generated PWM signal (V_{out}) on the oscilloscope. Record the peak-to-peak voltage (V_{pp}) and frequency of both signals.
5) Additionally, determine the duty cycle of the PWM signal by measuring the high-time (T_{high}) and low-time (T_{low}) of the waveform:

$$\text{Duty Cycle}(\%) = \frac{T_{high}}{T_{high} + T_{low}} \times 100.$$

6) Capture clear pictures of the oscilloscope display showing both the triangular wave and the PWM signal. Include these images in your report.
7) Write a discussion in your report analyzing the observed PWM signal. Explain how the duty cycle changes with the reference voltage and how this relates to the triangular wave input.
8) Summarize the role of the comparator in generating the PWM signal and the relationship between the input and output waveforms.

Pre-Report for Electronic Engineering III - Experiment 3: Waveform Generation using Operational

Group:
Date:

Student Identification

Matriculation Number	Student Name
2219941	Md Mostafizur Rahman
2220046	Ikramul Hasan Sazid

Pre-Report Tasks

- Perform simulations for Tasks 1, 2, and 3 using any appropriate software.
- Predict theoretical values for the parameters listed in the measurement table and fill in the corresponding column.
- The goal is to compare these theoretical predictions with the experimental results obtained during the lab session.
- Submit the pre-report before the experiment, attaching this page along with the completed simulation results.

Measurement Parameters by Task

TABLE IX: Measurement Parameters Organized by Task

Task	Parameter	Description	Measured Value
1	f	Expected frequency of the square wave.	50 Hz
1	V_{pp} of V_{out}	Peak-to-peak voltage of the output signal measured when $f = 25Hz$ (square wave).	30V
2	V_{pp} of V_{out}	Peak-to-peak voltage of the output signal measured when $f = 50Hz$ (triangular wave).	4V
3	V_{pp} of V_{out}	Peak-to-peak voltage of the output signal measured when $f = 50Hz$ (PWM).	30V
3	Duty Cycle (%)	Duty cycle of the PWM signal.	60%

Grades for Simulations

N_1	N_2	N_3	N_F

Notes:

- N_1: Grade for Task 1 simulation.
- N_2: Grade for Task 2 simulation.
- N_3: Grade for Task 3 simulation.
- N_F: Final grade, calculated as the average of the three tasks:

$$N_F = \frac{N_1 + N_2 + N_3}{3}$$